

Capital Structure Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Structure Discipline (CSD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CSD achieves an annualized gross (net) Sharpe ratio of 0.47 (0.41), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.45 (2.43), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 1.98.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting their capital stock and operations. While extensive research documents how investment frictions and financing constraints affect firm value, the precise mechanisms through which capital structure choices influence expected returns remain debated. We examine how Capital Structure Discipline (CSD) predicts cross-sectional variation in stock returns through the lens of production-based asset pricing. Firms exercising greater discipline in their capital structure decisions face different marginal costs of capital and adjustment frictions, creating systematic variation in their exposure to aggregate productivity shocks. This paper establishes that CSD serves as a powerful predictor of returns, generating risk-adjusted returns that cannot be explained by existing factor models or closely related anomalies.

In a production-based economy, firms' capital structure decisions directly affect their marginal costs of production and ability to respond to productivity shocks. Following the framework of [Cochrane and Piazzesi \(2005\)](#) and [Zhang \(2005\)](#), firms with disciplined capital structures maintain lower leverage and more stable financing, reducing their exposure to financial frictions that amplify adjustment costs. When productivity shocks hit the economy, these firms can more efficiently adjust their capital stock without facing prohibitive marginal financing costs. This flexibility translates into lower conditional risk premia, as investors demand less compensation for holding stocks of firms that can smoothly adjust to changing economic conditions.

The production-based theory of [Livdan et al. \(2009\)](#) demonstrates that financing frictions interact with real investment decisions to generate cross-sectional variation in expected returns. Firms exhibiting capital structure discipline effectively lower their marginal costs of external finance, enabling them to pursue positive NPV projects even during periods of aggregate uncertainty. This operational flexibility reduces their systematic risk exposure, as documented by [Chen and Novy-Marx \(2011\)](#),

who show that operating leverage amplifies firms' exposure to aggregate shocks. The CSD signal thus captures firms' differential exposure to productivity risk through the channel of financing constraints and adjustment costs.

Furthermore, [Berk et al. \(1999\)](#) establish that endogenous firm decisions regarding capital structure create persistent differences in risk profiles across firms. Companies maintaining disciplined capital structures commit to conservative financing policies that preserve financial flexibility, effectively purchasing real options to adjust production when faced with technology shocks. This commitment mechanism, as formalized by [Gomes and Schmid \(2010\)](#), generates predictable variation in conditional betas with respect to aggregate productivity factors. The CSD measure therefore proxies for firms' exposure to systematic investment frictions that vary with the business cycle, creating the cross-sectional return predictability we document.

The CSD trading strategy delivers economically and statistically significant returns across multiple specifications and robustness tests. A value-weighted long-short portfolio based on quintile sorts using NYSE breakpoints generates an average monthly return of 30 basis points with a t-statistic of 3.71, corresponding to an annualized Sharpe ratio of 0.47. The strategy's alpha remains highly significant after controlling for the Fama-French six factors, earning 19 basis points per month with a t-statistic of 2.45. These results demonstrate that CSD captures a distinct source of systematic risk not spanned by existing factor models.

The economic significance of CSD persists after accounting for transaction costs and across different portfolio construction methodologies. Net of trading costs, the strategy maintains a Sharpe ratio of 0.41 and generates significant alphas ranging from 14 to 44 basis points depending on the factor model specification. Notably, the CSD effect remains strong among large-cap stocks, where the strategy earns 24 basis points per month with a t-statistic of 2.64 for firms above the 80th NYSE size percentile. This finding rules out explanations based purely on limits to arbitrage or

market microstructure effects that plague many anomalies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the CSD strategy retains an alpha of 15 basis points per month with a t-statistic of 1.98. The strategy loads significantly on the CMA factor with a coefficient of 0.37 (t-statistic of 6.92), consistent with the interpretation that CSD captures firms' differential exposure to investment frictions. When combined with 154 existing anomalies using the average rank method, including CSD increases the terminal value of a dollar invested from \$2,560 to \$2,826, demonstrating its incremental contribution to the investment opportunity set.

This paper makes three distinct contributions to the asset pricing literature. First, we provide novel evidence that capital structure discipline predicts returns through a production-based channel, extending the work of [Fama and French \(2006\)](#) on the relationship between financing decisions and expected returns. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns and [Pontiff and Woodgate \(2008\)](#) show that share issuance contains pricing information, we demonstrate that CSD captures a broader dimension of financial discipline that subsumes these effects while maintaining independent predictive power. Our results complement [Lyandres and Palazzo \(2015\)](#) by showing that conservative capital structure policies proxy for lower exposure to systematic productivity risk.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the CSD anomaly across multiple dimensions. Unlike many documented anomalies that fail to survive transaction costs or are concentrated in small stocks, CSD generates significant net returns even among large-cap stocks and after controlling for microstructure effects. This finding addresses concerns raised by [Hou et al. \(2020\)](#) about the replicability of cross-sectional return predictors. Our spanning tests against 210 anomalies from the factor zoo demonstrate that CSD contains unique information about expected returns not

captured by existing predictors.

The broader implications of our findings extend to both academic research and practical portfolio management. For researchers, the CSD signal provides a parsimonious measure of firms' exposure to investment frictions that can be incorporated into structural asset pricing models. The strong performance of CSD in production-based frameworks supports recent theoretical advances by [Kogan and Papanikolaou \(2013\)](#) linking firm-level financing decisions to aggregate risk factors. For practitioners, the CSD strategy offers an implementable trading strategy that expands the mean-variance efficient frontier even after accounting for transaction costs and controlling for established factors, providing a valuable addition to multi-factor investment strategies.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital structure changes and to align with the typical frequency of major financing decisions. Capital Structure Discipline signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or retire treasury stock. Property, plant, and equipment gross (PPEGT) captures the total historical cost of tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and other physical capital investments. This measure provides a scale-

invariant denominator that reflects the firm’s cumulative investment in productive assets. We construct the Capital Structure Discipline signal by computing the year-over-year change in common stock scaled by lagged gross PPE: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{PPEGT}(t-1)$. This ratio captures the magnitude of equity financing activities relative to the firm’s existing capital base. Positive values indicate net equity issuance, while negative values suggest share repurchases or retirements, both scaled by the firm’s physical asset base. We calculate this measure using fiscal year-end values and require non-missing observations for both CSTK and PPEGT in consecutive years. Firms with zero or negative PPEGT in the lagged period are excluded from the analysis to avoid undefined or economically meaningless ratios. The resulting signal provides a standardized measure of how aggressively firms adjust their equity capital relative to their productive capacity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CSD signal. Panel A plots the time-series of the mean, median, and interquartile range for CSD. On average, the cross-sectional mean (median) CSD is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input CSD data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the CSD signal for the CRSP universe. On average, the CSD signal is available for 5.73% of CRSP names, which on average make up 7.10% of total market capitalization.

4 Does CSD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CSD using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CSD portfolio and sells the low CSD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CSD strategy earns an average return of 0.30% per month with a t-statistic of 3.71. The annualized Sharpe ratio of the strategy is 0.47. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.37, with a t-statistic of 6.92 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 482 stocks and an average market capitalization of at least \$1,352 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.63. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CSD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the CSD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CSD, as well as average returns and alphas for long/short trading CSD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CSD strategy achieves an average return of 24 bps/month

with a t-statistic of 2.64. Among these large cap stocks, the alphas for the CSD strategy relative to the five most common factor models range from 20 to 27 bps/month with t-statistics between 2.11 and 2.90.

5 How does CSD perform relative to the zoo?

Figure 2 puts the performance of CSD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CSD strategy falls in the distribution. The CSD strategy’s gross (net) Sharpe ratio of 0.47 (0.41) is greater than 87% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CSD strategy (red line).² Ignoring trading costs, a \$1 invested in the CSD strategy would have yielded \$6.77 which ranks the CSD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CSD strategy would have yielded \$4.85 which ranks the CSD strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CSD relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the CSD strategy gross alphas fall between the 61 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CSD strategy has a positive net generalized alpha for five out of the five factor models. In these cases CSD ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CSD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CSD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CSD or at least to weaken the power CSD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CSD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CSD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CSD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CSD. Stocks are finally grouped into five CSD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CSD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CSD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CSD signal in these Fama-MacBeth regressions exceed 1.71, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on CSD is 1.19.

Similarly, Table 5 reports results from spanning tests that regress returns to the CSD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CSD strategy earns alphas that range from 13-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.71, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CSD trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.98.

7 Does CSD add relative to the whole zoo?

Finally, we can ask how much adding CSD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the CSD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes CSD grows to \$2826.29.

8 Conclusion

This study provides compelling evidence that Capital Structure Discipline (CSD) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CSD maintains economically and statistically significant predictive power even after accounting for transaction costs and controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CSD generates robust risk-adjusted returns, with an annualized net Sharpe ratio of 0.41

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CSD is available.

and statistically significant alphas relative to the Fama-French five-factor model plus momentum. Notably, the signal’s persistence is evident in its ability to deliver a monthly alpha of 15 basis points (t -statistic = 1.98) even after controlling for six closely related strategies from the factor zoo, including net payout yield, share issuance measures, and asset growth variables. This suggests that CSD captures unique information about firm value that is not fully subsumed by existing factors related to corporate financing decisions and capital allocation.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, CSD adds to our understanding of how managerial decisions regarding capital structure influence expected returns, supporting theories that link financial discipline to firm value creation. For practitioners, the signal’s ability to generate positive net returns after transaction costs suggests potential implementation value, though the moderate Sharpe ratio indicates it may be most effective when combined with other signals in a multi-factor framework.

Several limitations of this study warrant consideration. First, while we control for numerous related factors, the possibility remains that CSD proxies for an omitted risk factor or behavioral bias not captured in our analysis. Second, the implementation assumptions, while following established protocols, may not fully reflect real-world trading constraints, particularly for less liquid securities. Third, the statistical significance of the alpha relative to the extended factor model (t -statistic = 1.98) is marginal, suggesting that the incremental information content of CSD, while present, may be modest.

Future research should explore several promising directions. First, investigating the economic mechanisms underlying the CSD premium would enhance our theoretical understanding of why capital structure discipline predicts returns. Second, examining the signal’s performance across different market conditions, particularly

during periods of financial stress, could reveal important time-variation in its effectiveness. Third, international evidence would help establish whether the CSD effect represents a global phenomenon or is specific to certain market structures. Finally, exploring potential enhancements to the CSD measure, perhaps by incorporating forward-looking information or industry-specific adjustments, may improve its predictive power. Understanding these aspects will be crucial for determining whether Capital Structure Discipline represents a persistent market anomaly or a risk factor that should be incorporated into asset pricing models.

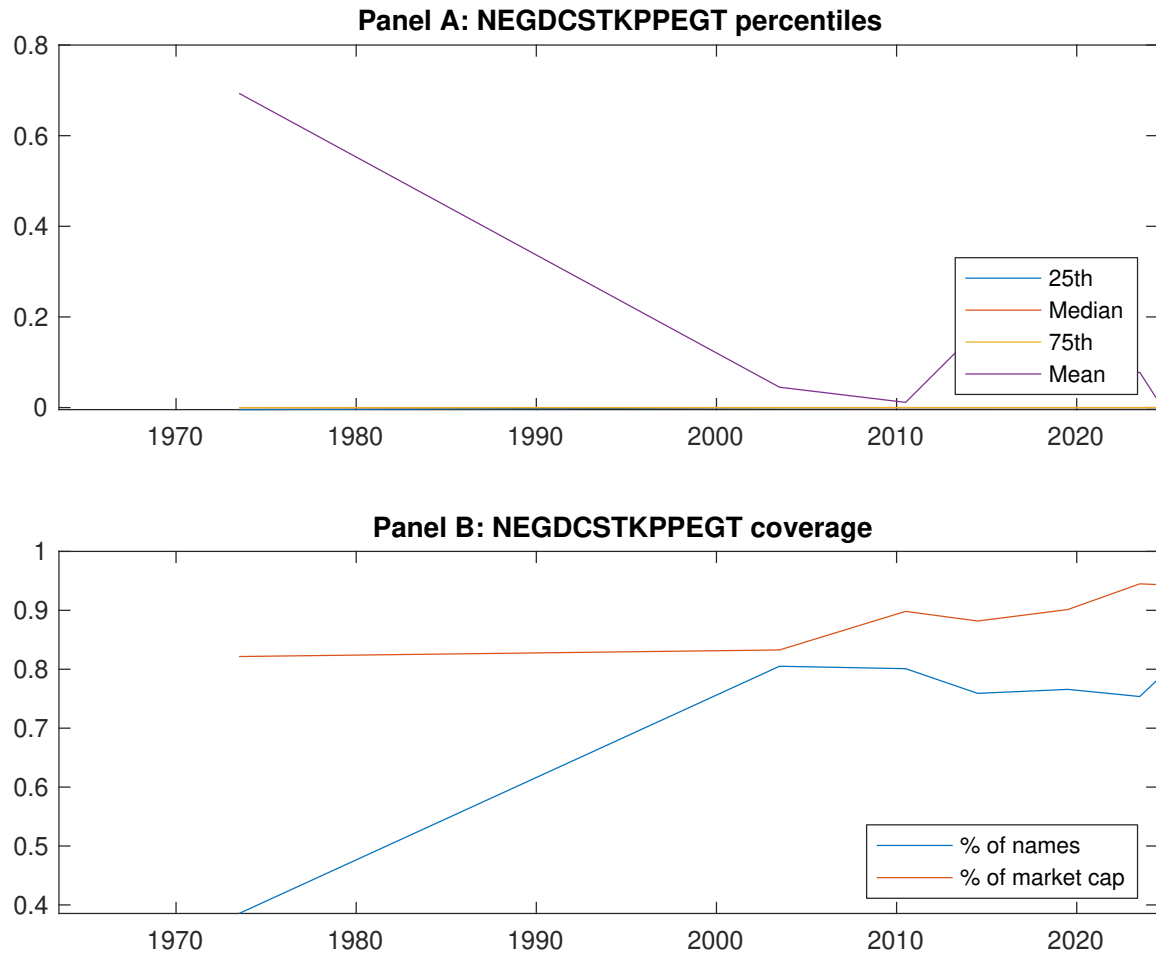


Figure 1: Times series of CSD percentiles and coverage. This figure plots descriptive statistics for CSD. Panel A shows cross-sectional percentiles of CSD over the sample. Panel B plots the monthly coverage of CSD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CSD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on CSD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.65]	0.53 [2.97]	0.66 [3.76]	0.64 [4.04]	0.76 [4.74]	0.30 [3.71]
α_{CAPM}	-0.13 [-2.59]	-0.08 [-1.79]	0.06 [1.20]	0.10 [2.13]	0.22 [4.45]	0.35 [4.30]
α_{FF3}	-0.09 [-1.86]	-0.05 [-1.17]	0.08 [1.68]	0.08 [1.70]	0.19 [3.99]	0.28 [3.61]
α_{FF4}	-0.08 [-1.60]	-0.02 [-0.36]	0.08 [1.67]	0.06 [1.23]	0.18 [3.59]	0.26 [3.20]
α_{FF5}	-0.10 [-2.07]	0.01 [0.14]	0.06 [1.26]	-0.00 [-0.03]	0.10 [2.22]	0.21 [2.66]
α_{FF6}	-0.10 [-1.87]	0.03 [0.74]	0.07 [1.30]	-0.01 [-0.24]	0.10 [2.09]	0.19 [2.45]
Panel B: Fama and French (2018) 6-factor model loadings for CSD-sorted portfolios						
β_{MKT}	0.98 [80.57]	1.00 [92.29]	1.01 [82.52]	0.97 [89.40]	0.98 [86.31]	-0.00 [-0.13]
β_{SMB}	0.03 [1.62]	0.04 [2.28]	0.02 [1.38]	-0.08 [-4.84]	-0.02 [-0.94]	-0.04 [-1.61]
β_{HML}	-0.07 [-3.00]	-0.09 [-4.41]	-0.09 [-3.73]	-0.00 [-0.21]	-0.04 [-1.89]	0.03 [0.79]
β_{RMW}	0.10 [4.05]	-0.13 [-6.20]	0.04 [1.74]	0.10 [4.52]	0.11 [4.76]	0.01 [0.24]
β_{CMA}	-0.11 [-3.07]	-0.04 [-1.42]	0.03 [0.92]	0.22 [7.25]	0.27 [8.27]	0.37 [6.92]
β_{UMD}	-0.01 [-1.04]	-0.04 [-3.59]	-0.00 [-0.33]	0.01 [1.23]	0.01 [0.57]	0.02 [1.01]
Panel C: Average number of firms (n) and market capitalization (me)						
n	732	605	482	591	663	
me (\$10 ⁶)	1619	1352	1879	2065	2365	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CSD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [3.71]	0.35 [4.30]	0.28 [3.61]	0.26 [3.20]	0.21 [2.66]	0.19 [2.45]
Quintile	NYSE	EW	0.52 [6.74]	0.62 [8.50]	0.51 [8.05]	0.42 [6.69]	0.34 [5.78]	0.28 [4.74]
Quintile	Name	VW	0.26 [3.25]	0.30 [3.78]	0.24 [3.04]	0.21 [2.61]	0.17 [2.13]	0.15 [1.90]
Quintile	Cap	VW	0.24 [3.10]	0.28 [3.65]	0.24 [3.11]	0.20 [2.61]	0.19 [2.48]	0.17 [2.16]
Decile	NYSE	VW	0.32 [3.34]	0.38 [3.93]	0.28 [3.05]	0.24 [2.53]	0.27 [2.92]	0.24 [2.53]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.23]	0.31 [3.86]	0.25 [3.19]	0.23 [2.97]	0.19 [2.54]	0.19 [2.43]
Quintile	NYSE	EW	0.32 [3.85]	0.44 [5.50]	0.33 [4.77]	0.28 [4.14]	0.17 [2.66]	0.14 [2.26]
Quintile	Name	VW	0.22 [2.75]	0.26 [3.30]	0.20 [2.59]	0.18 [2.35]	0.15 [1.96]	0.14 [1.83]
Quintile	Cap	VW	0.20 [2.63]	0.25 [3.18]	0.20 [2.64]	0.18 [2.37]	0.17 [2.28]	0.16 [2.11]
Decile	NYSE	VW	0.28 [2.88]	0.32 [3.38]	0.24 [2.57]	0.21 [2.29]	0.23 [2.52]	0.21 [2.34]

Table 3: Conditional sort on size and CSD

This table presents results for conditional double sorts on size and CSD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CSD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CSD and short stocks with low CSD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CSD Quintiles					CSD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.45 [1.59]	0.71 [2.69]	0.85 [3.23]	0.99 [4.01]	1.01 [4.30]	0.58 [5.35]	0.68 [6.34]	0.57 [5.91]	0.50 [5.17]	0.36 [3.95]	0.31 [3.46]
	(2)	0.49 [1.98]	0.76 [3.19]	0.89 [3.68]	0.91 [4.05]	0.95 [4.31]	0.46 [4.58]	0.55 [5.59]	0.41 [4.65]	0.35 [3.90]	0.32 [3.53]	0.27 [3.00]
	(3)	0.65 [2.97]	0.59 [2.79]	0.77 [3.49]	0.79 [3.88]	0.96 [4.89]	0.31 [3.45]	0.39 [4.34]	0.28 [3.38]	0.24 [2.84]	0.22 [2.53]	0.19 [2.17]
	(4)	0.52 [2.62]	0.62 [3.06]	0.81 [4.05]	0.75 [3.98]	0.81 [4.45]	0.28 [3.18]	0.35 [3.96]	0.23 [2.98]	0.23 [2.92]	0.04 [0.52]	0.06 [0.75]
	(5)	0.47 [2.83]	0.54 [3.06]	0.50 [2.92]	0.52 [3.18]	0.72 [4.54]	0.24 [2.64]	0.27 [2.90]	0.23 [2.47]	0.20 [2.15]	0.21 [2.32]	0.20 [2.11]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CSD Quintiles					CSD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	342	342	342	340	335	26	29	33	26	24	
	(2)	97	95	96	96	95	50	49	51	49	49	
	(3)	69	69	69	69	69	87	85	87	88	89	
	(4)	60	60	60	60	60	188	189	198	195	199	
(5)	56	56	56	56	55	1324	1387	1487	1563	1734		

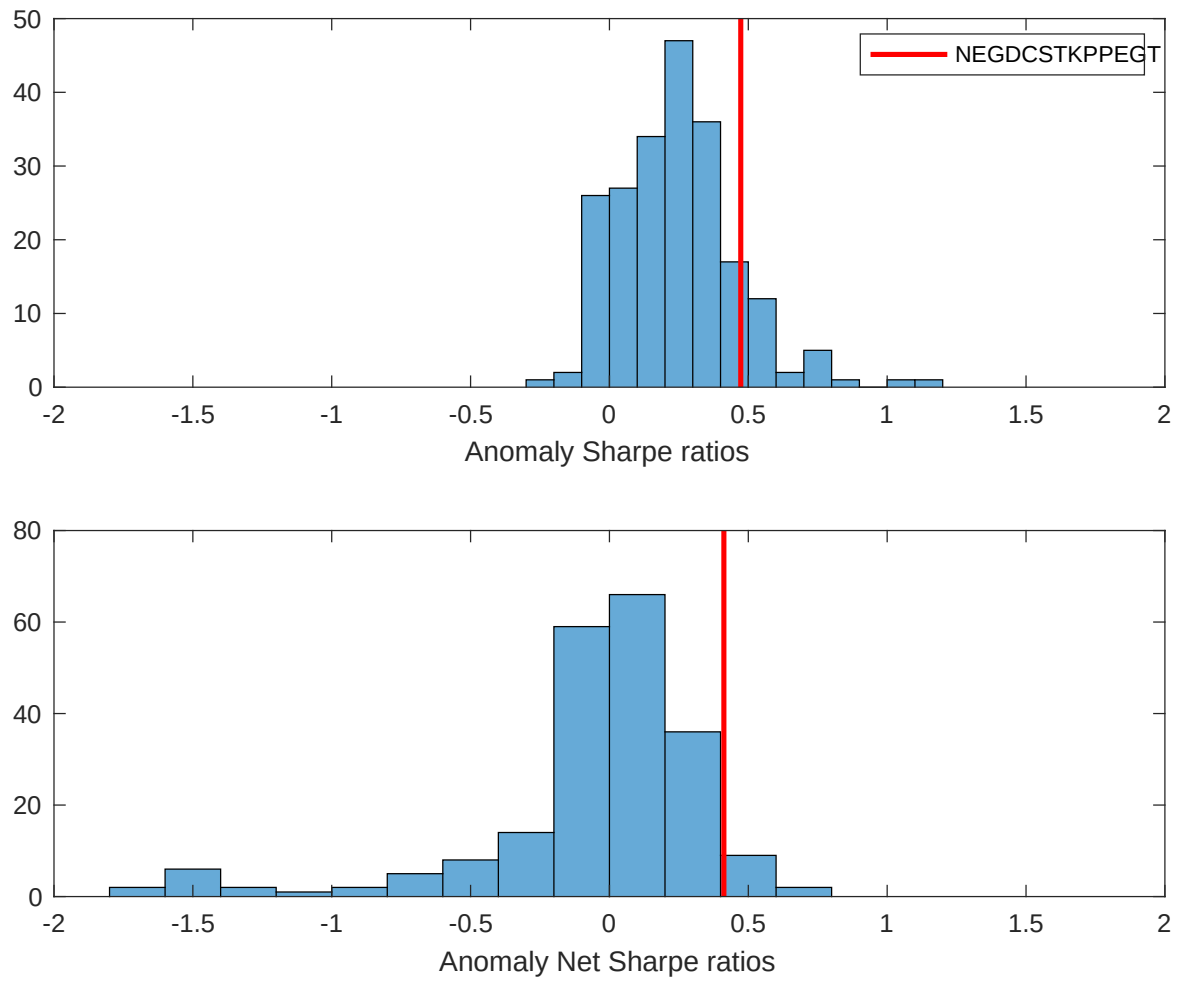


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CSD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

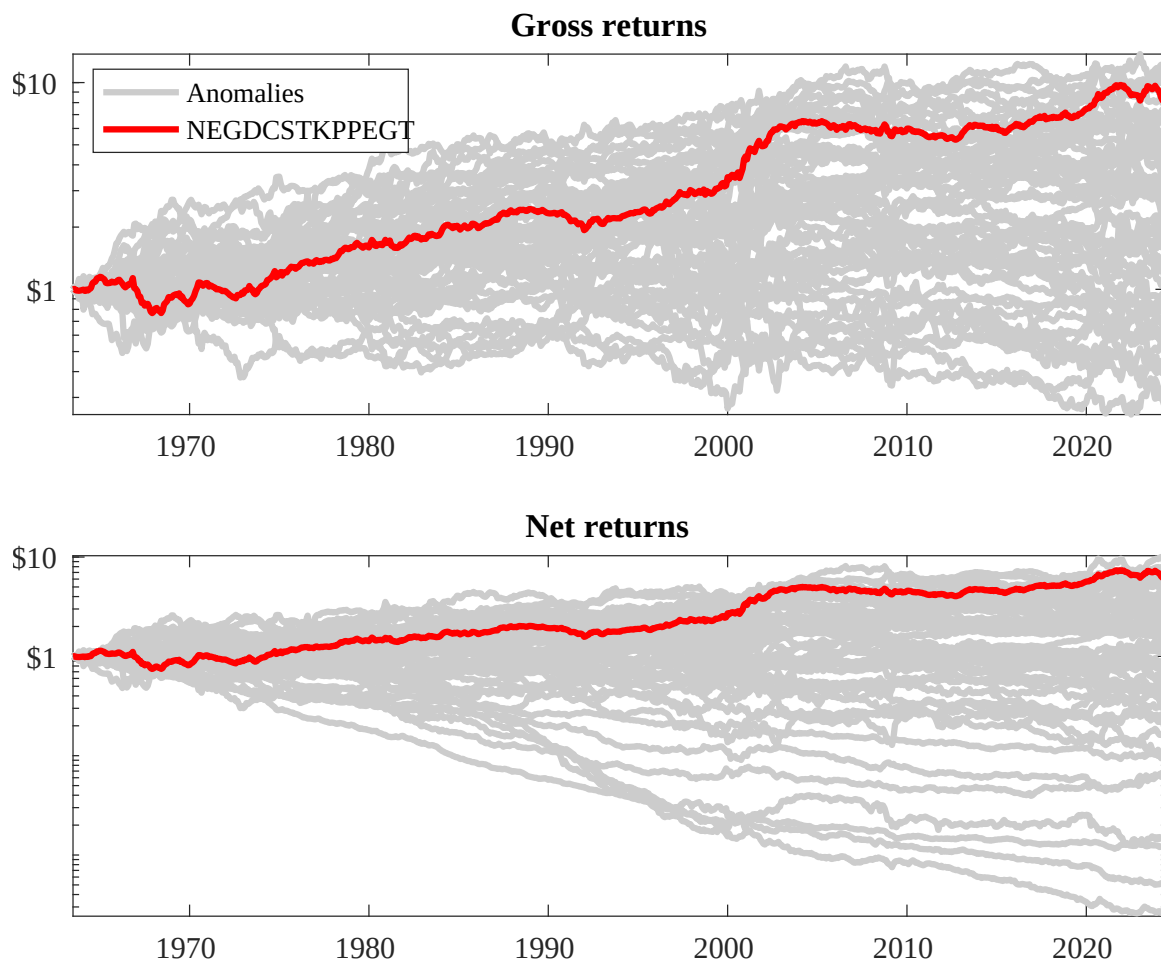
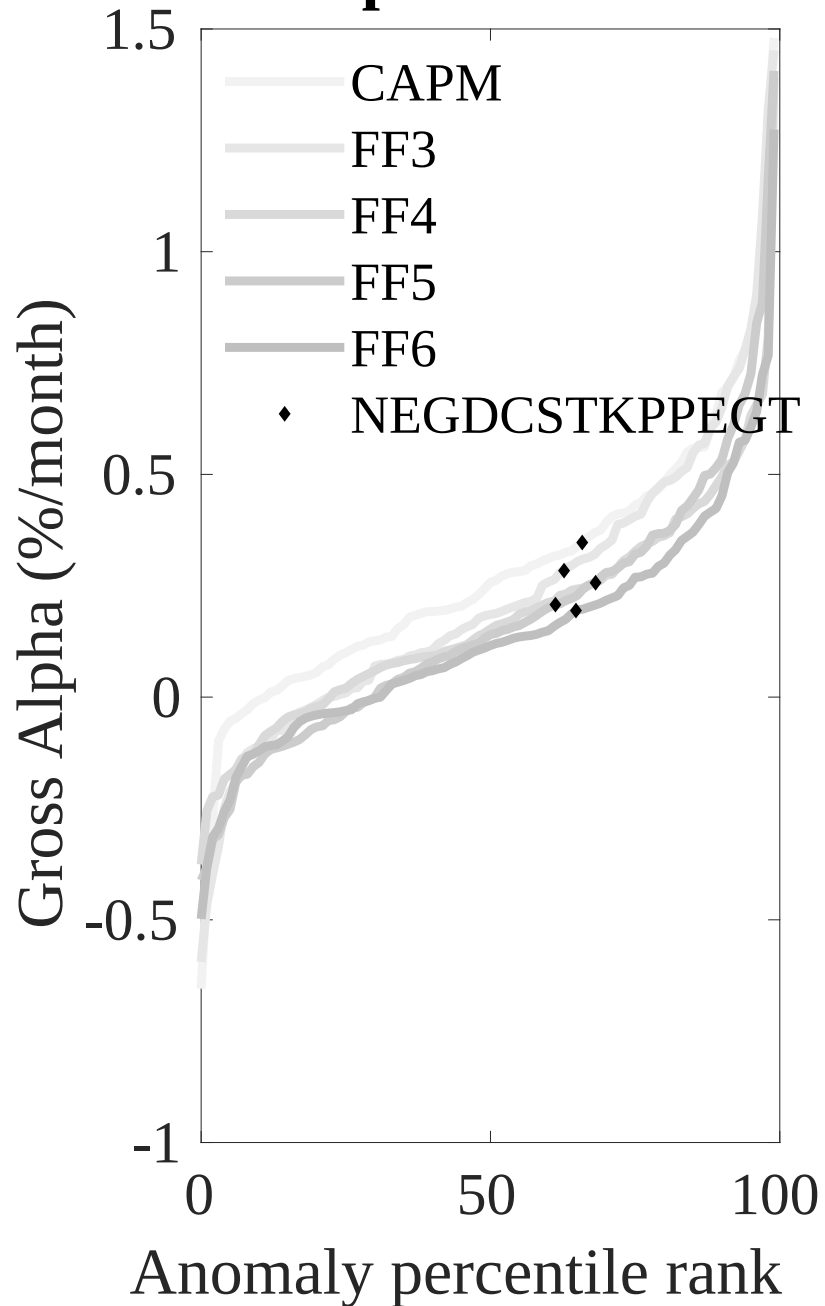


Figure 3: Dollar invested.

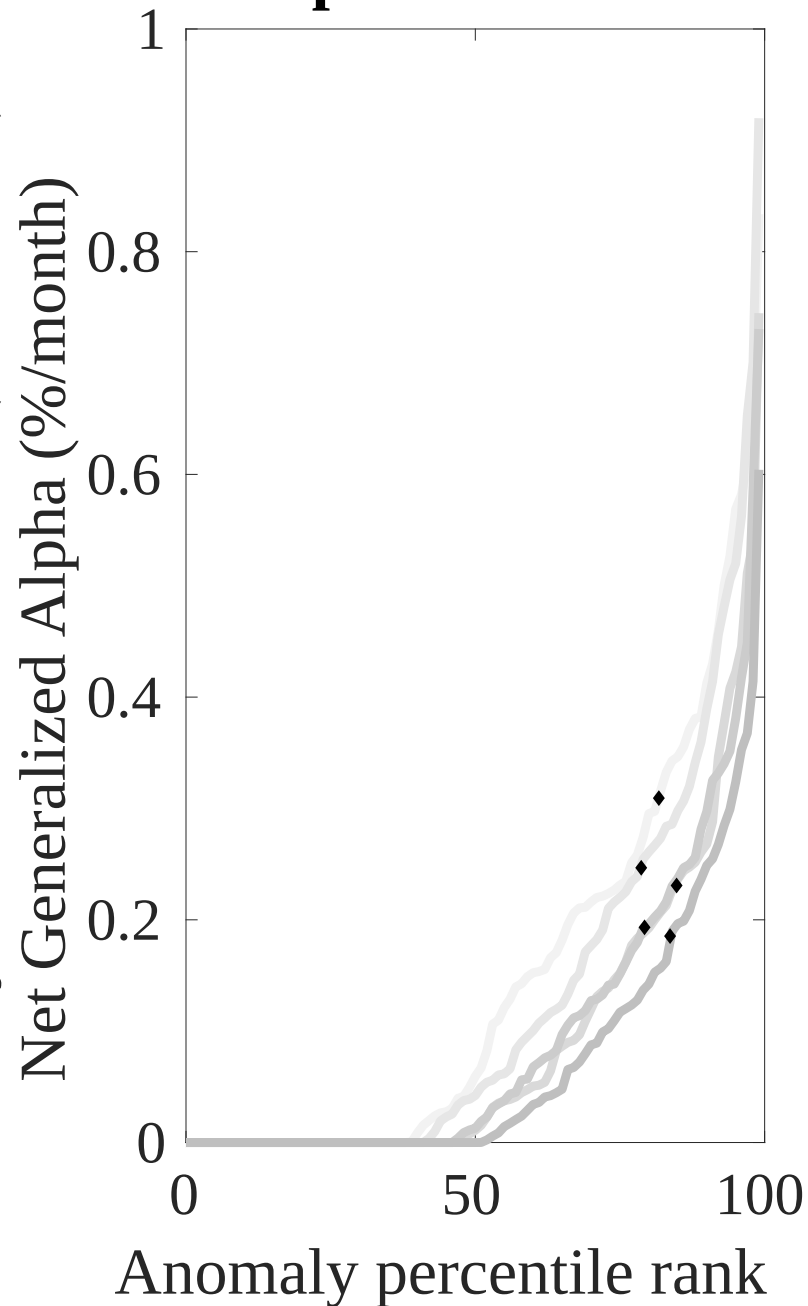
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CSD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



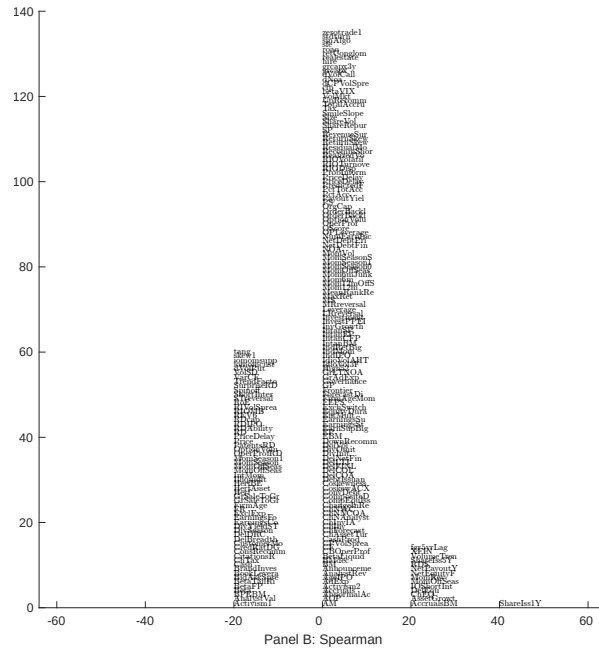
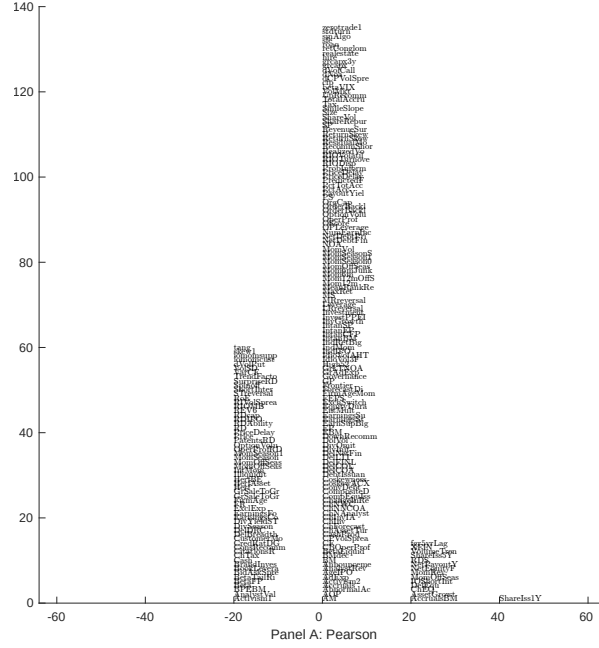


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CSD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

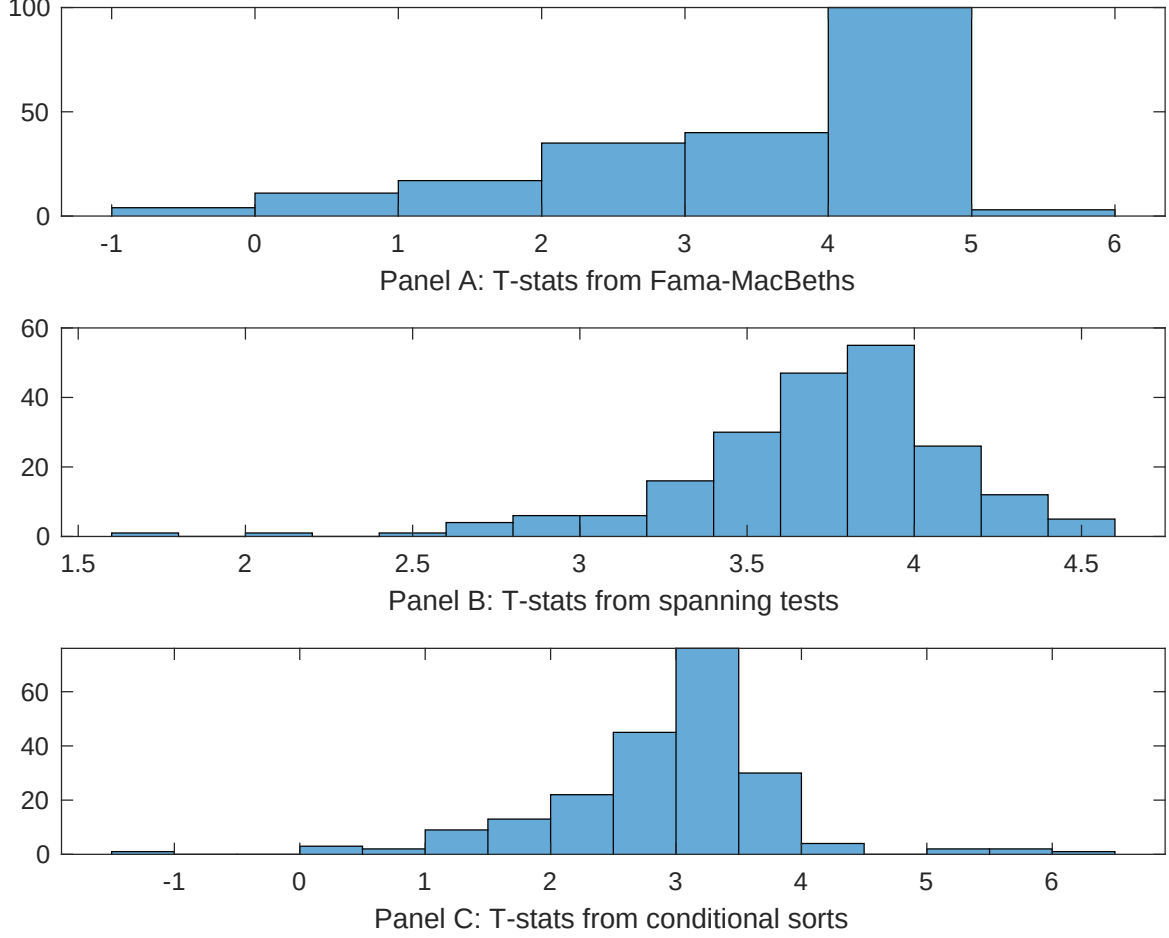


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CSD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CSD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CSD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CSD. Stocks are finally grouped into five CSD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CSD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CSD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CSD}CSD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.42]	0.13 [5.97]	0.17 [6.69]	0.13 [6.36]	0.13 [5.59]	0.13 [6.02]	0.12 [3.87]
CSD	0.88 [1.71]	0.17 [3.57]	0.16 [3.13]	0.16 [3.39]	0.19 [3.74]	0.15 [3.05]	0.61 [1.19]
Anomaly 1	0.22 [1.96]						1.00 [0.94]
Anomaly 2		0.14 [6.18]					0.68 [2.48]
Anomaly 3			0.39 [3.32]				-0.15 [-0.65]
Anomaly 4				0.28 [4.85]			0.24 [4.23]
Anomaly 5					0.12 [3.26]		-0.24 [-0.04]
Anomaly 6						0.93 [7.89]	0.64 [5.39]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	1	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CSD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CSD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net Payout Yield, Share issuance (1 year), Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23 [2.96]	0.18 [2.36]	0.18 [2.36]	0.13 [1.71]	0.19 [2.49]	0.19 [2.41]	0.15 [1.98]
Anomaly 1	18.85 [6.35]						8.64 [2.67]
Anomaly 2		25.82 [6.67]					10.24 [2.21]
Anomaly 3			31.97 [7.82]				38.06 [6.44]
Anomaly 4				25.66 [6.26]			11.70 [2.54]
Anomaly 5					13.38 [3.46]		-10.85 [-2.00]
Anomaly 6						-3.05 [-0.65]	-19.69 [-3.87]
mkt	2.87 [1.53]	0.60 [0.33]	1.01 [0.56]	1.12 [0.61]	-0.28 [-0.15]	-0.04 [-0.02]	3.45 [1.91]
smb	-0.15 [-0.05]	-2.22 [-0.84]	-4.22 [-1.62]	-1.53 [-0.58]	-3.32 [-1.23]	-2.90 [-1.06]	-0.07 [-0.03]
hml	-3.15 [-0.84]	3.58 [1.02]	0.35 [0.10]	6.32 [1.79]	2.52 [0.69]	4.44 [1.23]	-1.23 [-0.34]
rmw	-10.09 [-2.54]	-13.07 [-3.17]	2.41 [0.69]	-8.28 [-2.15]	2.12 [0.58]	0.94 [0.26]	-12.88 [-3.00]
cma	20.55 [3.59]	20.34 [3.58]	5.47 [0.85]	23.92 [4.32]	22.97 [3.50]	40.13 [5.17]	15.14 [1.95]
umd	3.59 [1.97]	0.91 [0.50]	1.54 [0.86]	0.42 [0.23]	2.44 [1.32]	2.04 [1.10]	-0.00 [-0.00]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	20	20	22	19	16	15	27

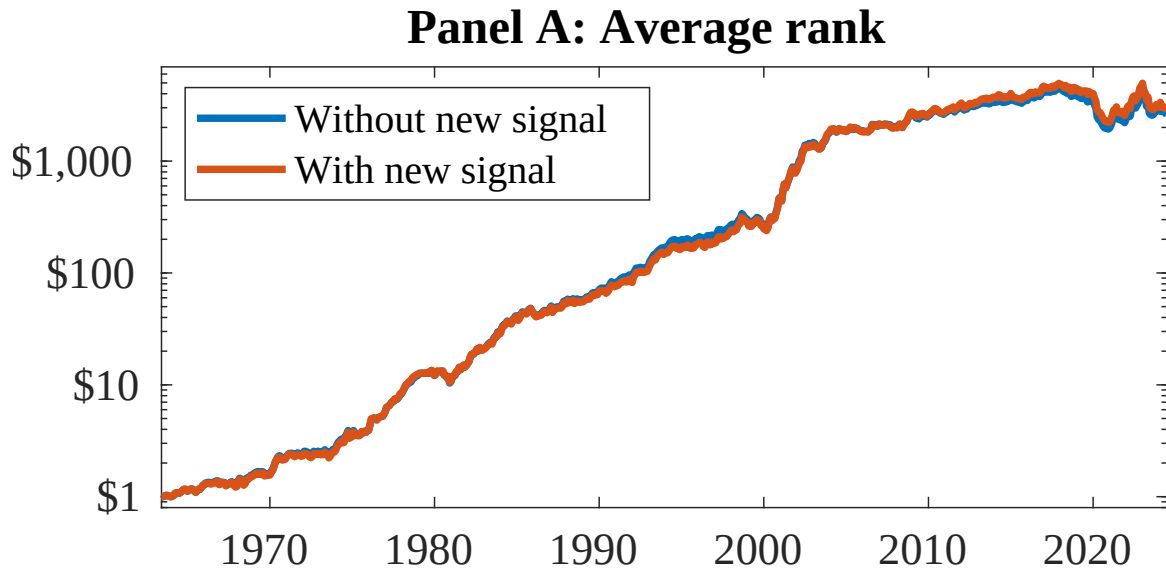


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as CSD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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